

AI 內容學習路線 Cantonese-English Revision & Quiz

38 份 PDF 閱讀次序、10 週實作計劃、粵語速解、English examples、每個主題 Quiz 同獨立答案

Covers 1,290 pages | 8-10 hours per week | Bilingual revision edition |
2026-09-03

呢本唔係第 39 份只讀唔練嘅資料

Every topic now follows Learn -> Explain -> Example -> Quiz -> Review. 每個 Quiz 都先獨立做，答案集中放近書尾；錯題要變成 Day 1、Day 3、Day 7 嘅重溫卡。源 PDF 保持不變。

先做這件事：安全清零

P0 風險

部分項目材料中疑似存在 API Key、AK/SK 或追蹤平台密鑰樣式的字串。若其中任何憑據仍有效，應立即吊銷並輪換；不要把憑據複製到代碼、筆記、日誌或新的 PDF。Kubernetes Secret 的 Base64 也不是加密。

開始學習前的 60 分鐘檢查

- 輪換所有可能暴露的憑據；在倉庫啟用 secret scanning，並檢查 Git 歷史。
- 只使用合成數據或已獲明確授權的數據；微信聊天、客服錄音、截圖、音色復刻都需要用途、保存期限、訪問範圍和刪除機制。
- 把所有示例密鑰替換為佔位符；通過環境變量或密鑰管理服務注入，日誌必須脫敏。
- 區分“已實現、已驗證、設計方案、後續計劃”；沒有測試證據的性能數字只能寫成目標或示例。
- 刪除任何鼓勵虛構公司、工作年限、團隊、營收、日活或上線事故的求職建議。

為什麼放在最前面

資料中的技術示例可能被直接複製。安全、隱私和誠信錯誤一旦進入倉庫、簡歷或生產環境，後續再補救通常比重新寫一份示例昂貴得多。

怎麼讀：掃 - 畫 - 跑 - 改 - 講

每份 PDF 都按同一循環處理。閱讀不是終點，能解釋、能復現、能測量才算完成。

步驟	時間	動作	留下的證據
掃	10 分鐘	看目錄、標題、圖表，先寫 3 個問題。	問題清單與閱讀範圍
畫	15 分鐘	畫輸入、輸出、狀態、依賴和失敗點。	一張系統圖或流程圖
跑	30-60 分鐘	在鎖定版本的環境中跑最小示例。	可重複命令與運行結果
改	20 分鐘	只改變一個變量，並測量差異。	小型對照實驗
講	5 分鐘	關掉資料，用自己的話解釋；第 1、3、7 天溫習。	口頭錄音或回憶卡

每份文檔的一頁學習卡

- 一句話主旨。
- 一張圖：數據流、狀態機或調用鏈。
- 三個核心概念。
- 兩個失敗模式。
- 一個有數字的實驗。
- 三個閉卷問題。

時間分配

預設每週 8-10 小時：約 60% 動手、25% 閱讀、15% 回憶與復盤。若只閱讀沒有交付，下週不要繼續擴展新主題。

每個主題有咩 | What each topic includes

1 learning-plan page + 1 Cantonese-English revision page + 1 closed-book quiz page. The answer key is deliberately kept near the end, so you can test recall before checking. 每題做錯都要寫低原因，唔係淨係圈返正確答案。

路線總覽：10 周建立一條能力主線

周	主題	關鍵產出
1	AI 基礎與 Python	環境、JSON CLI、FastAPI /ask
2	LLM 原理與 API	多模式 API 包裝器與 20 例基準
3	RAG 與 GraphRAG	30 問檢索評測與參數對照
4	Agent 與 MCP	只讀 Agent、本地 MCP、審批邊界
5	LangChain	三工具 Agent、結構化輸出、10 個測試
6	LangGraph 與 LangSmith	可恢復工作流、追蹤與回歸評測
7	微調與數據工程	授權數據集、LoRA 實驗、盲測
8	數據與電商項目	數據血緣、ETL、審計、回滾
9	客服、銷售與內容項目	一個可部署的縱向切片
10	簡歷與項目面試	證據化簡歷、項目陳述、模擬面試

能力依賴

Python	>	LLM/API	>	RAG	>	Agent/MCP	>	框架
數據工程	>	微調	>	項目	>	評測	>	作品集

必要前置

- 會使用 Python 函數、集合、異常、文件與基礎 async。
- 能發送 HTTP 請求、解析 JSON，並完成 PostgreSQL 基礎 CRUD。
- 會使用 Git，能創建隔離環境並通過 lockfile 復現依賴。
- GPU 只在第 7 週可能需要；也可以用小型雲任務或記錄完整的 dry run。

01

AI 基礎與 Python

先把後端、API、數據與運行環境說清楚

閱讀順序

- 0基礎到Ai銜接部分.pdf
- 新版-懂王Ai-Python從入小白到精通.pdf

本週要掌握

- 解釋瀏覽器、API、後端、數據庫與模型之間的數據流。
- 使用函數、類型提示、異常、文件、async 與 HTTP 客戶端。
- 用 pyproject.toml 與 lockfile 固定可重複環境。

動手交付

- 實現一個 JSON 清洗 CLI。
- 實現 FastAPI /ask：輸入 JSON，調用一個可替換的模型適配器，返回結構化 JSON。
- 為正常、超時、無效輸入與取消請求寫測試。

退出標準

- 不用看筆記，能畫出前端到模型的完整調用鏈。
- 在新目錄中用一條命令重建環境並通過測試。

校正提醒

Python 是動態且強類型語言；參數傳遞更準確的說法是“按賦值/對象共享”。線程並非“假的”：I/O 場景仍有價值，且自由線程構建已經出現。不要把某個 Python 小版本永遠稱為“當前版本”。

R01

AI 與 Python 基礎 / AI and Python Foundations

Cantonese recap, English explanation, example and key vocabulary

粵語速解 | Cantonese recap

學 AI 應用唔係一開始就訓練大模型。先睇清流程：前端收集輸入，經 API 交俾 Python 後端；後端驗證資料、處理業務邏輯、呼叫 LLM，再用 JSON 將結果送返前端。Python 適合做呢層，因為 FastAPI、資料處理同 AI SDK 生態成熟。實作時要用 uv 或 Conda 隔離環境，金鑰放喺環境變數，唔好提交去 Git。真正嘅入門標準，係可以由零開一個服務、處理錯誤、測試輸入輸出，亦清楚講得出每一層負責做乜。

English explanation

An AI application is usually a normal software system with an AI service inside it. The frontend collects input, the backend validates it and applies business rules, and an API connects the backend to an LLM. Python is popular because FastAPI makes API development straightforward and the AI ecosystem is mature.

Example | 例子

A browser sends an order number to FastAPI. The backend validates it, retrieves the order, optionally asks an LLM to phrase the reply, and returns JSON such as status: shipped. The API key stays on the server and never appears in the browser response.

Key vocabulary | 重點英文詞彙

English term	粵語意思 / Cantonese meaning
Frontend	用戶睇到同操作嘅介面
Backend	喺伺服器處理資料同邏輯嘅層
API	兩個系統溝通嘅入口同規格
JSON	常用嘅鍵值資料格式
Virtual environment	將每個專案嘅依賴隔離
Async I/O	等網絡時唔阻塞其他工作

Revision tip | 溫習貼士

遮住右邊解釋，只睇 English term，用廣東話講一次意思，再用英文講一個例子。If you cannot explain it both ways, mark it for Day 1, Day 3 and Day 7 review.

Q01

AI and Python Foundations Revision Quiz

Do it closed-book first. Answers are collected near the end of the guide.

做法：先唔好揭答案。MCQ 每題 1 分，短答每題 2 分，實作題 3 分；滿分 10 分。Try to answer the short questions in both Cantonese and English.

1. Where should a production LLM API key normally be stored?
 - A. In frontend JavaScript
 - B. In a backend environment variable
 - C. In a public Git repository
 - D. In the returned JSON
2. What is the main purpose of a virtual environment?
 - A. Make Python syntax shorter
 - B. Encrypt every source file
 - C. Isolate project dependencies
 - D. Replace an API server
3. What should a backend do before processing submitted JSON?
 - A. Validate its structure and values
 - B. Display its secret keys
 - C. Convert every field to text
 - D. Forward it without inspection

Short answer | 短答

4. Describe the request path from a browser to an LLM and back.

5. Give two security or reliability rules for a Python API.

Mini practical challenge | 小實作

6. Build POST /ask with a validated question, blank-input rejection, a mock environment secret, JSON output, and tests for one valid plus two invalid requests.

Your evidence / 你嘅證據：

Score guide

8-10: ready to move on. 6-7: review your weak items tomorrow. 0-5: re-run the smallest example before continuing.

02

LLM 原理與 API

把穩定概念與供應商特有介面分開

閱讀順序

- LLM大模型原理 (claude4.6 精簡版) .pdf
- LLM模型原理.pdf (作為深入查閱)
- 推理模型LLM Api (物料) .pdf

本週要掌握

- 理解 token、embedding、attention、FFN、殘差、歸一化、KV cache、量化與批處理。
- 區分 messages、tools、schema、streaming 等穩定概念與不同供應商的欄位差異。
- 理解 TTFT、總延遲、吞吐、Token 與失敗率。

動手交付

- 構建統一 API 包裝器，支持普通響應、流式、結構化 JSON 與圖片輸入。
- 用 20 個固定提示運行基準，記錄 TTFT、總延遲、Token、錯誤和 schema 有效率。
- 建立 provider/model capability matrix 與 smoke test。

退出標準

- 更換模型或供應商時，業務層無需重寫。
- 所有功能聲稱都有日期、模型版本與測試結果。

校正提醒

“OpenAI-compatible”是業界兼容叫法，不是 W3C 標準。top_p 是對累積概率質量進行 nucleus sampling，不是“控制生僻詞”。不要要求或展示原始思維鏈；使用可見摘要和可驗證結果。

R02

LLM 原理與 API / LLM Principles and APIs

Cantonese recap, English explanation, example and key vocabulary

粵語速解 | Cantonese recap

LLM 本質上係根據上文預測下一個 token 嘅模型。文字先經 tokenizer 拆成 token，再由 embedding 變成向量；Transformer 用自注意力分析各部分關係，逐步產生答案。訓練通常包括預訓練、SFT 同偏好對齊；API 推理只係用固定權重計算，唔會因為一次對話自動學識新知識。開發時要分清 system、user、assistant 訊息，控制 temperature 同輸出長度，並量度 TTFT、總延遲、token 同成本。結構化輸出仍要用 schema 驗證；串流只係早啲顯示內容，唔代表總運算一定更少。

English explanation

An LLM predicts the next token from the tokens already present in its context. A tokenizer converts text into tokens, embeddings turn them into vectors, and Transformer attention layers calculate relationships among them. During inference, trained weights remain fixed; a normal API conversation does not retrain the model.

Example | 例子

Given the prompt 'The capital of France is', the model gives high probability to 'Paris'. Streaming may display the first token sooner, but the complete generation can still require similar total computation. Always record model version, TTFT, total latency, token usage, and schema failures.

Key vocabulary | 重點英文詞彙

English term	粵語意思 / Cantonese meaning
Token	模型處理文字嘅細單位
Embedding	將內容變成帶語義嘅向量
Transformer	用注意力處理序列嘅架構
Self-attention	計算 token 之間嘅關係
Inference	用固定權重產生答案
Structured output	按 schema 返回可驗證資料

Revision tip | 溫習貼士

遮住右邊解釋，只睇 English term，用廣東話講一次意思，再用英文講一個例子。If you cannot explain it both ways, mark it for Day 1, Day 3 and Day 7 review.

Q02

LLM Principles and APIs Revision Quiz

Do it closed-book first. Answers are collected near the end of the guide.

做法：先唔好揭答案。MCQ 每題 1 分，短答每題 2 分，實作題 3 分；滿分 10 分。Try to answer the short questions in both Cantonese and English.

1. What normally happens to model weights during API inference?

- A. They are deleted
- B. They remain fixed
- C. They are retrained after every message
- D. They become part of the JSON response

2. Lowering temperature usually makes output what?

- A. More deterministic
- B. Longer in every case
- C. Factually perfect
- D. Independent of the prompt

3. Which approach is most reliable for machine-readable output?

- A. Ask politely for JSON
- B. Split arbitrary prose
- C. Increase temperature
- D. Enforce a schema and validate the result

Short answer | 短答

4. Distinguish pre-training, supervised fine-tuning, and inference.

5. Explain the difference between TTFT and total latency.

Mini practical challenge | 小實作

6. Create an LLM wrapper or mock that supports normal and streaming responses, validates a JSON schema, and measures ten fixed prompts.

Your evidence / 你嘅證據：

Score guide

8-10: ready to move on. 6-7: review your weak items tomorrow. 0-5: re-run the smallest example before continuing.

03

RAG 與知識圖譜

先測檢索，再討論模型回答

閱讀順序

- Rag架構、向量原理.pdf
- LangChain案例.pdf
- 你非懂不可GraphRag.pdf

本週要掌握

- 掌握切片、embedding、dense/sparse hybrid、BM25、rerank 與 ACL 過濾。
- 區分兩步 RAG、agentic RAG 與 GraphRAG 的適用場景。
- 理解索引版本、刪除、重建、來源追蹤與引用準確率。

動手交付

- 為一組授權文檔構建 RAG，準備 30 個問題與參考證據。
- 比較 chunk 128/256/512 與 top-k 3/5；報告 Recall@k、MRR/nDCG、答案忠實度和引用準確率。
- 畫出 GraphRAG local/global/DRIFT 查詢的選擇圖。

退出標準

- 可以用數據解釋參數選擇，而不是引用固定“黃金參數”。
- 刪除或撤回文檔後，索引與引用同步更新。

校正提醒

embedding 維度更大不等於必然更准，top-k 也沒有通用常數。向量數據庫並非強制；現有 SQL/搜索系統能滿足評測目標時可以繼續使用。檢索到的文本應被當作不可信輸入。

R03

RAG 與 GraphRAG / RAG and GraphRAG

Cantonese recap, English explanation, example and key vocabulary

粵語速解 | Cantonese recap

RAG 係先搵資料，再俾 LLM 回答，唔會改模型權重。離線時將文件切成 chunk，用 embedding 轉成向量，再存入索引；查詢時亦將問題向量化，召回相關片段，組成有來源嘅 prompt。效果唔只睇模型，chunk 大細、overlap、top-k、metadata filter 同 reranker 都會影響準確度。GraphRAG 會再抽取實體同關係，建立圖結構：local query 適合問特定人物或事件，global query 適合分析整體模式。佢擅長多跳關係，但建圖成本高，抽取錯誤亦可能沿住張圖擴散，所以唔係每個知識庫都需要用。

English explanation

RAG supplements an LLM with retrieved external evidence without changing its weights. Documents are chunked, embedded, and indexed. At query time, the system retrieves relevant chunks, optionally reranks them, and places the best evidence into the prompt. GraphRAG adds entities and relationships for multi-hop questions.

Example | 例子

For 'When is salary paid?', ordinary RAG can retrieve one handbook paragraph saying 'on the 15th'. For 'How are Alice, Project Apollo, and the security team connected?', GraphRAG may be useful because the answer follows several relationships rather than one paragraph.

Key vocabulary | 重點英文詞彙

English term	粵語意思 / Cantonese meaning
Chunking	將長文件切成語義片段
Vector database	儲存向量兼做相似搜尋
Similarity search	按距離或夾角搵相關內容
Reranker	對初步召回結果再精排
Knowledge graph	用節點同邊表示實體關係
GraphRAG	結合圖檢索同 LLM 生成

Revision tip | 溫習貼士

遮住右邊解釋，只睇 English term，用廣東話講一次意思，再用英文講一個例子。If you cannot explain it both ways, mark it for Day 1, Day 3 and Day 7 review.

Q03

RAG and GraphRAG Revision Quiz

Do it closed-book first. Answers are collected near the end of the guide.

做法：先唔好揭答案。MCQ 每題 1 分，短答每題 2 分，實作題 3 分；滿分 10 分。Try to answer the short questions in both Cantonese and English.

1. Why can excessively large chunks reduce retrieval precision?

- A. They always delete the answer
- B. They retrain the LLM
- C. They add unrelated context and dilute relevance
- D. They disable JSON

2. What is the main job of a reranker?

- A. Generate training embeddings
- B. Reorder retrieved candidates with stronger scoring
- C. Replace the LLM
- D. Encrypt the knowledge base

3. Which question is the strongest GraphRAG candidate?

- A. What is the exact refund deadline?
- B. Show paragraph three
- C. What is today's date?
- D. How are these people, teams, and projects connected?

Short answer | 短答

4. List a sensible diagnostic sequence when relevant documents are not being retrieved.

5. Contrast local and global queries in GraphRAG.

Mini practical challenge | 小實作

6. Build a 12-passage RAG baseline for ten questions, compare two chunk sizes and two top-k values, then sketch when a graph extension would add value.

Your evidence / 你嘅證據：

Score guide

8-10: ready to move on. 6-7: review your weak items tomorrow. 0-5: re-run the smallest example before continuing.

04

Agent 與 MCP

先建立權限邊界，再增加自主性

閱讀順序

- Ai Agent全貌概覽.pdf
- Agent集群架構 副本.pdf
- 深入淺出DeepAgent.pdf

本週要掌握

- 區分 workflow 的預定路徑與 agent 的動態決策。
- 理解 tool/resource/prompt、host/client/server 與傳輸邊界。
- 掌握預算、停止條件、冚等、重試、超時、工具白名單與人工批准。

動手交付

- 實現一個只讀三工具 Agent。
- 實現一個本地 MCP server/client，並畫出認證、配額、trace ID、共享記憶和審批點。
- 對提示注入、越權工具、惡意工具結果和超預算循環做測試。

退出標準

- 寫操作預設需要審批，工具按 read/write/destructive 分類。
- 任何一次工具調用都可追蹤、可重試或安全終止。

校正提醒

MCP 2026-07-28 已採用無狀態核心，移除了舊的 initialize/initialized 與 Mcp-Session-Id，並加入 server/discover、MRTR、路由頭和擴展框架。舊教程必須注明協議版本；遠程授權還要做最小權限、受眾綁定和令牌隔離。

R04

Agent 與 MCP / Agents and MCP

Cantonese recap, English explanation, example and key vocabulary

粵語速解 | Cantonese recap

Agent 唔只係識傾偈；佢會按目標揀工具、睇結果，再決定下一步。可預測工作應先用固定流程，需要判斷先交界 Agent。MCP 係連接工具同資料嘅開放協議。2026-07-28 版改成無狀態核心，取消初始化握手同 Session ID，加入可選 server/discover、MRTR 同 HTTP 路由標頭。實務上要最小權限、分清讀寫工具、寫入前批核，並設成本、逾時、重試同停止條件。工具輸出只係外來資料，可能過時、出錯或藏有提示注入，唔可以當可信指令；真正權限控制一定要由程式同伺服器執行。

English explanation

An agent is a control loop that observes state, selects an action, uses a tool, and decides whether to continue. It should not replace a deterministic workflow when the sequence is known. MCP standardizes connections between an AI host and servers exposing tools, resources, and prompts. In MCP 2026-07-28, requests are self-describing and the core protocol is stateless.

Example | 例子

An expense agent may call `read_policy` and `calculate_total` automatically, but `submit_reimbursement` requires human approval. The host enforces that rule outside the prompt, uses an idempotency key, caps the run at eight tool calls, and logs every write. MRTR can request a missing cost centre without keeping a bidirectional session open.

Key vocabulary | 重點英文詞彙

English term	粵語意思 / Cantonese meaning
Agent loop	觀察、揀行動、用工具、再判斷嘅循環
Deterministic workflow	步驟預先定好、結果較易預測嘅流程
MCP	連接 AI 主機、工具同資料來源嘅開放協議
MRTR	工具中途需要資料時，以多次請求完成工作
Least privilege	每個工具只攞完成任務所需嘅最低權限
Idempotency	同一寫入重試都唔會重複造成副作用

Revision tip | 溫習貼士

遮住右邊解釋，只睇 English term，用廣東話講一次意思，再用英文講一個例子。If you cannot explain it both ways, mark it for Day 1, Day 3 and Day 7 review.

Q04

Agents and MCP Revision Quiz

Do it closed-book first. Answers are collected near the end of the guide.

做法：先唔好揭答案。MCQ 每題 1 分，短答每題 2 分，實作題 3 分；滿分 10 分。Try to answer the short questions in both Cantonese and English.

- Which statement matches MCP 2026-07-28?
 - Every client must initialize a session
 - Each request can be self-describing, with optional server/discover
 - Mcp-Session-Id is mandatory
 - Legacy HTTP+SSE is the preferred new transport
- What is the safest treatment of a payment-submission tool?
 - Let the model call it at high confidence
 - Add 'be careful' to the prompt
 - Enforce authorization and approval outside the model
 - Trust metadata without verification
- When should a deterministic workflow normally be preferred?
 - When the tool list is long
 - When context is large
 - When every task is open-ended
 - When the sequence and rules are already known

Short answer | 短答

- Why must an agent treat tool output as untrusted data?

- How does MRTR support a tool that needs user input midway?

Mini practical challenge | 小實作

- Design a purchasing agent with search_catalog, draft_order, and submit_order, including approval, retries, budgets, audit logs, and malicious supplier text handling.

Your evidence / 你嘅證據：

Score guide

8-10: ready to move on. 6-7: review your weak items tomorrow. 0-5: re-run the smallest example before continuing.

05

LangChain

學習框架角色，而不是背 API

閱讀順序

- (新版2.0) 深入淺出LangChain.pdf (主教材)
- LangChain全解析.pdf (配套)
- 深入淺出LangChain.pdf (版本對照)

本週要掌握

- 使用高層 Agent 抽象、tools、middleware、runtime context、memory 與 structured output。
- 理解 LCEL 的組合價值，但不把舊寫法當成唯一主線。
- 用可替換介面隔離框架與供應商。

動手交付

- 構建一個包含 3 個工具的 Agent，支持結構化輸出、流式與持久記憶。
- 固定依賴版本，編寫 10 個可重複測試，包括工具錯誤與超時。
- 比較一次直接 SDK 實現與框架實現的複雜度、可觀測性和維護成本。

退出標準

- 能夠說明什麼時候不需要 LangChain。
- 升級依賴後通過契約測試發現行為差異。

校正提醒

框架示例必須記錄版本。LangChain 是高層 Agent 框架；LangGraph 是低層 orchestration runtime；Deep Agents 更接近帶計劃、子 Agent、文件系統與上下文管理的 harness。

R05

LangChain 應用 / LangChain Applications

Cantonese recap, English explanation, example and key vocabulary

粵語速解 | Cantonese recap

LangChain v1 嘅主線係用 `create_agent` 組合模型、工具同 `middleware`，而唔係死背一大堆舊 Chain。Middleware 可以動態改提示、揀模型、重試、處理工具錯誤、摘要歷史、遮走私隱資料同加入人工批核。需要固定格式時，可以用 `ProviderStrategy` 或 `ToolStrategy`，最後結果會放喺 `structured_response`。短期記憶通常靠 `checkpoint` 同 `thread_id`；`create_agent` 底層由 `LangGraph` 執行。LCEL 仍然適合用管道砌細型資料流程，但唔代表每個應用都要變成 Agent。無論用邊種方法，都要驗證工具參數、限制權限，並測試失敗、逾時同格式錯誤。

English explanation

LangChain v1 uses `create_agent` as the standard high-level entry point. A model, tools, response schema, runtime context, and middleware are assembled into an agent that runs on `LangGraph`. Middleware handles concerns such as model selection, prompt changes, PII redaction, retries, tool errors, summarization, and approval.

Example | 例子

A support agent receives an authenticated `customer_id` through runtime context, calls a read-only `lookup_order` tool, and returns a validated `SupportReply`. Middleware redacts card numbers and requires approval before exposing `refund_order`. A simple load-transform-classify pipeline may still use LCEL without agent autonomy.

Key vocabulary | 重點英文詞彙

English term	粵語意思 / Cantonese meaning
<code>create_agent</code>	LangChain v1 建立 Agent 嘅標準入口
Middleware	喺模型或工具前後加入治理邏輯
Tool schema	定義同驗證工具參數及結果嘅規格
Runtime context	執行時注入、唔一定畀模型睇嘅資料
Structured output	按指定 schema 產生可驗證結果
LCEL	用 Runnable 同管道組合資料處理步驟

Revision tip | 溫習貼士

遮住右邊解釋，只睇 English term，用廣東話講一次意思，再用英文講一個例子。If you cannot explain it both ways, mark it for Day 1, Day 3 and Day 7 review.

Q05

LangChain Applications Revision Quiz

Do it closed-book first. Answers are collected near the end of the guide.

做法：先唔好揭答案。MCQ 每題 1 分，短答每題 2 分，實作題 3 分；滿分 10 分。Try to answer the short questions in both Cantonese and English.

- What is the recommended LangChain v1 entry point for a standard agent?
 - create_agent
 - LLMChain
 - ConversationChain
 - A legacy prebuilt ReAct helper
- Where does create_agent place its validated structured result?
 - messages[0]
 - structured_response
 - tool_calls[-1]
 - runtime.context
- Which mechanism best fits PII redaction, dynamic prompts, and tool-error handling?
 - A larger context window
 - Middleware
 - A vector database
 - Higher temperature

Short answer | 短答

- Compare ProviderStrategy with ToolStrategy.

- Give two examples of data that belongs in runtime context rather than model-visible messages.

Mini practical challenge | 小實作

- Sketch a LangChain v1 order-support agent with lookup, typed SupportReply, card-number redaction, and approval before refund.

Your evidence / 你嘅證據：

Score guide

8-10: ready to move on. 6-7: review your weak items tomorrow. 0-5: re-run the smallest example before continuing.

06

LangGraph 與 LangSmith

讓長流程可恢復、可觀測、可評測

閱讀順序

- (2.0新版) 深入淺出LangGraph.pdf (主教材)
- LangGraph全解析.pdf
- LangSmith鏈路追蹤監控.pdf

本週要掌握

- 掌握 state、node、edge、checkpoint、interrupt/resume、subgraph 與 streaming。
- 區分 tracing、offline evaluation 與 online evaluation。
- 理解 replay-safe node 與副作用冪等。

動手交付

- 構建包含路由、並行節點、checkpoint、interrupt/resume 與 subgraph 的工作流。
- 運行中強制重啟並證明恢復；記錄重複執行的副作用檢查。
- 把失敗 trace 加入離線數據集，做版本回歸對比。

退出標準

- 一次失敗能夠定位到 node、模型、工具或數據層。
- 部署前後的評測使用同一份版本化數據集。

校正提醒

LangGraph 在 interrupt 後恢復時會從節點開頭重新執行，interrupt 前的副作用必須冪等，或拆到獨立節點。Checkpoint 的具體表結構屬於後端實現細節，不應寫成穩定協議。

R06

LangGraph 與 LangSmith / LangGraph and LangSmith

Cantonese recap, English explanation, example and key vocabulary

粵語速解 | Cantonese recap

LangGraph 適合將多步 AI 工作寫成有狀態圖：節點負責做事、邊控制流向、checkpointer 保存進度，而 thread_id 指向同一段工作。最易出錯係 interrupt：恢復時唔係由停低嗰行繼續，而係成個節點由頭再跑，所以批核前嘅副作用要做到冚等，或者搬去下一個節點。LangSmith 就唔只係睇 trace；開發時應用離線資料集做比較、回歸同元件測試，上線後再用線上評估監察真實流量。發現失敗個案，就去除私隱資料、人工核對，再加入離線測試集，形成可以持續改進嘅閉環。

English explanation

LangGraph represents an application as state, nodes, and transitions. A checkpointer persists state, while thread_id identifies the run to resume. Resuming an interrupt restarts its node from the beginning, so external writes before it must be idempotent or moved to a separate post-approval node. LangSmith supports tracing plus offline and online evaluation.

Example | 例子

A refund graph uses validate_order, request_approval, then issue_refund. The approval node only prepares a proposal and pauses. After resumption, issue_refund uses an idempotency key. LangSmith tests routing and tool arguments offline, then monitors unsafe refunds, excessive steps, latency, and failures online.

Key vocabulary | 重點英文詞彙

English term	粵語意思 / Cantonese meaning
StateGraph	用狀態、節點同邊表示工作流程
Checkpointer	保存圖狀態，令工作可以恢復
thread_id	指向同一段持久化工作嘅識別碼
Interrupt	暫停圖並等候人或外部輸入
Trace	記錄模型、工具、步驟、時間同錯誤
Evaluator	按準則量度輸出或執行軌跡質素

Revision tip | 溫習貼士

遮住右邊解釋，只睇 English term，用廣東話講一次意思，再用英文講一個例子。If you cannot explain it both ways, mark it for Day 1, Day 3 and Day 7 review.

Q06

LangGraph and LangSmith Revision Quiz

Do it closed-book first. Answers are collected near the end of the guide.

做法：先唔好揭答案。MCQ 每題 1 分，短答每題 2 分，實作題 3 分；滿分 10 分。Try to answer the short questions in both Cantonese and English.

1. What happens when a LangGraph node resumes after interrupt?

- A. It continues from the next source line
- B. The entire graph restarts at START
- C. The interrupted node restarts from its beginning
- D. The checkpoint skips all earlier code

2. What is the relationship between a checkpoint and thread_id?

- A. They select model and temperature
- B. The checkpoint stores state and thread_id identifies which state to load
- C. They are billing fields
- D. They replace tool-call IDs

3. Which evaluation pairing is correct?

- A. Offline for regression; online for production monitoring
- B. Offline for live traffic only
- C. Online only before deployment
- D. There is no practical difference

Short answer | 短答

4. How should a payment or refund side effect be placed around an interrupt?

5. Describe a loop that turns a production failure into a regression test.

Mini practical challenge | 小實作

6. Design a human-approved refund graph with state, nodes, interrupt payload, thread_id, idempotency, and two offline plus two online evaluation criteria.

Your evidence / 你嘅證據：

Score guide

8-10: ready to move on. 6-7: review your weak items tomorrow. 0-5: re-run the smallest example before continuing.

07

模型微調與數據工程

先證明 Prompt 與 RAG 不夠，再訓練

閱讀順序

- 數據標注平台搭建與實戰 副本.pdf
- 主流模型微調方案解讀 副本.pdf
- LLaMA-Factory+Lora模型微調實戰 副本.pdf
- 項目3：聊天數據微調-完整流程文檔版本 副本.pdf

本週要掌握

- 掌握數據授權、去標識、版本、血緣、會話級切分與防洩漏。
- 理解 SFT、LoRA/QLoRA、preference data、DPO 等用途。
- 把 rank、alpha、learning rate 與數據比例當作待實驗假設。

動手交付

- 用授權數據導出 conversational、prompt-completion 與 preference 格式。
- 按用戶/會話/時間切分 train/validation/test，再做樣本擴展。
- 運行一個小型 LoRA/QLoRA 實驗，與 base+prompt+RAG 在同一凍結測試集上盲測。

退出標準

- 能說明為什麼必須微調，以及它相對簡單基線的收益。
- 記錄數據 hash、模型 revision、seed、包版本、硬件、成本和失敗案例。

校正提醒

“70% 數據/30% 算法”、alpha=2x rank、固定 2e-4 學習率等只能當作實驗起點。4-bit 權重大小不等於訓練顯存：還要計算 activation、adapter、optimizer、metadata、kernel 與 runtime overhead。

R07

微調與數據工程 / Fine-tuning and Data Engineering

Cantonese recap, English explanation, example and key vocabulary

粵語速解 | Cantonese recap

微調唔應該係第一步：先試清晰提示；知識成日更新就用 RAG；只有穩定嘅語氣、格式、分類或工具行為先考慮 SFT。偏好資料可以用 DPO，其他方法亦要靠實驗決定，唔好見新就追。資料工程要先取得同意、處理私隱同授權，並按完整對話、客戶或時間切分，再展開成訓練樣本，避免資料洩漏。訓練時要用模型本身嘅 chat template，檢查特殊 token、遮罩、截斷同 assistant-only loss。LoRA 或 QLoRA 可以慳記憶體，但估算時仲要計 activation、adapter、optimizer 同運行開銷。最後一定要同提示及 RAG 基線比較。

English explanation

Fine-tuning is useful when a measured, stable behavior cannot be achieved reliably through prompting or retrieval. Use RAG for changing knowledge and SFT for behavior, style, formatting, classification, or tool-use demonstrations. Preference methods such as DPO require meaningful chosen and rejected pairs and should be adopted only when evaluation shows a gain.

Example | 例子

For 10,000 support conversations, obtain permission, remove PII, deduplicate, and split by whole conversation or customer before expanding samples. Apply the model's chat template, train a LoRA baseline, and compare it with prompt-only and RAG systems on task success, format, tone, privacy leakage, latency, and cost. Four-bit weight size is not total training memory.

Key vocabulary | 重點英文詞彙

English term	粵語意思 / Cantonese meaning
SFT	用示範輸入同理想輸出做監督式微調
LoRA	只訓練低秩 adapter，減少可訓練參數
QLoRA	量化基礎模型，再訓練 LoRA adapter
Chat template	將角色訊息轉成模型預期格式
Data leakage	測試資料或相關內容滲入訓練資料
Preference optimization	用偏好比較令模型傾向較理想答案

Revision tip | 溫習貼士

遮住右邊解釋，只睇 English term，用廣東話講一次意思，再用英文講一個例子。If you cannot explain it both ways, mark it for Day 1, Day 3 and Day 7 review.

Q07

Fine-tuning and Data Engineering Revision Quiz

Do it closed-book first. Answers are collected near the end of the guide.

做法：先唔好揭答案。MCQ 每題 1 分，短答每題 2 分，實作題 3 分；滿分 10 分。Try to answer the short questions in both Cantonese and English.

1. A product catalogue changes every day. What is normally the best first approach?

- A. Full model pretraining
- B. RAG over the current catalogue
- C. DPO on last year's answers
- D. Increase LoRA rank

2. Why should conversations be split before expansion into multiple samples?

- A. Increase GPU use
- B. Remove tokenization
- C. Prevent related turns leaking across train and evaluation sets
- D. Guarantee preference labels

3. What does a four-bit raw-weight estimate omit?

- A. Nothing
- B. Only disk compression
- C. Only the evaluation data
- D. Activations, adapters, optimizer state, kernels, metadata, and runtime overhead

Short answer | 短答

4. What should be recorded to make a fine-tuning run reproducible?

5. When might DPO be justified after an SFT baseline?

Mini practical challenge | 小實作

6. Plan a pipeline for 10,000 support chats covering consent, PII, deduplication, splitting, chat templates, baselines, LoRA training, evaluation, and provenance.

Your evidence / 你嘅證據：

Score guide

8-10: ready to move on. 6-7: review your weak items tomorrow. 0-5: re-run the smallest example before continuing.

08

數據、素材與電商項目

把數據血緣、權限與回滾做成主角

閱讀順序

- 包裝電商項目需要下去瞭解的內容.pdf (掃讀並補業務流程)
- 項目6：Ai數據中台.pdf
- 項目8：Ai素材管理中心.pdf
- 項目9 商品交易自進化Agent -- Harness與Loop Engineer 架構設計.pdf

本週要掌握

- 設計 raw -> staging -> approved -> RAG/SFT/assets 的數據血緣。
- 為文字、圖片、影片與用戶反饋建立授權、ACL、審計和刪除流程。
- 理解自進化只應產生候選策略，不能直接修改線上規則。

動手交付

- 建立 schema、一個真實 ETL 路徑、審計歷史、回滾與數據品質檢查。
- 為跨租戶洩漏、RAG 投毒、重複任務與失敗恢復寫測試。
- 把策略更新放在離線評測、人工批准、版本發佈與可回滾流程之後。

退出標準

- 任意輸出可以追溯到數據來源、處理版本和批准人。
- 刪除請求能覆蓋原始數據、索引、緩存與訓練派生物。

校正提醒

“已脫敏”不等於獲得訓練授權。對話、錄音、截圖、音色和用戶畫像必須有明確用途與刪除機制。銷售評分不得成為員工獎懲的唯一依據，應提供復核與申訴。

R08

數據、素材庫與電商專案 / Data, Assets and E-commerce Projects

Cantonese recap, English explanation, example and key vocabulary

粵語速解 | Cantonese recap

做數據中台、素材庫同電商 AI 專案，唔好一開始就堆模型。先畫清楚資料由邊度嚟、邊個有權用、保存幾耐。穩陣流程係 raw -> staging -> approved：原始層供追溯，暫存層做去重、脫敏同人工審核，只有獲批資料先入 RAG 或微調。圖片同影片要保存有授權嘅描述、標籤、SKU 同 retrieval text；查詢時先做租戶、權限同 SKU 硬篩選，再做語義檢索。API key 絕對唔可以放喺 PDF、程式碼或日誌。最後用 Recall@K、跨租戶洩漏率、素材命中率同每次成功查詢成本驗收；未量度嘅數字只可以叫目標。

English explanation

A strong data or e-commerce AI project begins with governance, not a vector database. Separate raw, review, and approved data so transformations remain traceable and only authorized records reach production retrieval or training. Exact permissions and product identifiers should filter candidates before semantic ranking.

Example | 例子

For 'Does the red X2 headset have a two-year warranty?', identify X2-RED, enforce tenant and SKU permissions, retrieve approved product text and media, and answer from the current warranty record. Low confidence triggers clarification or human escalation. Customer chats need consent and retention controls; credentials come from a secret manager.

Key vocabulary | 重點英文詞彙

English term	粵語意思 / Cantonese meaning
Data lineage	記錄資料由來源去到輸出嘅路徑
Staging layer	等待清洗同人工審核嘅暫存層
Data minimisation	只收集完成目的所需嘅最少資料
Retrieval text	代表圖片或影片、供語義搜尋嘅文字
Tenant isolation	防止唔同客戶互相睇到資料
Idempotency	同一操作重試都唔會重複執行

Revision tip | 溫習貼士

遮住右邊解釋，只睇 English term，用廣東話講一次意思，再用英文講一個例子。If you cannot explain it both ways, mark it for Day 1, Day 3 and Day 7 review.

Q08

Data, Assets and E-commerce Projects Revision Quiz

Do it closed-book first. Answers are collected near the end of the guide.

做法：先唔好揭答案。MCQ 每題 1 分，短答每題 2 分，實作題 3 分；滿分 10 分。Try to answer the short questions in both Cantonese and English.

1. Which record is safest to place in a production RAG index?
- A. Every raw customer message
 - B. An approved, authorized, policy-compliant record
 - C. Any record scored highly by an LLM
 - D. A screenshot found online
2. A query contains an exact SKU. What should happen before semantic ranking?
- A. Remove the SKU
 - B. Search every tenant
 - C. Apply tenant, permission, and SKU filters
 - D. Fine-tune the model
3. A live API key appears in a PDF. What is the correct first response?
- A. Hide the filename
 - B. Delete only the visible line
 - C. Revoke and rotate it, then inspect history and logs
 - D. Encode it with Base64

Short answer | 短答

4. Why should a raw layer be traceable but still follow retention and deletion rules?

5. Name two tests that help detect cross-tenant leakage in RAG.

Mini practical challenge | 小實作

6. Design a pipeline for ten product SKUs with FAQs, images, and videos, including states, access controls, secrets, retrieval order, and three metrics.

Your evidence / 你嘅證據：

Score guide

8-10: ready to move on. 6-7: review your weak items tomorrow. 0-5: re-run the smallest example before continuing.

09

客服、銷售與內容項目

選一個縱向切片做到可部署

閱讀順序

- 項目一：Ai語音對話-智能客服項目 副本.pdf
- 項目二：自媒體運營Ai智能體 副本.pdf
- 項目5：直播切片Agent 項目 .pdf
- 項目7：銷售考核Agent.pdf

本週要掌握

- 從業務目標反推 SLI/SLO、測試集、成本預算和降級策略。
- 使用 trace/metric/log 串聯 API、隊列、工具、數據庫與模型調用。
- 明確 MVP 與生產版邊界，記錄 ADR 與 failure matrix。

動手交付

- 在語音/RAG 流式或 LangGraph 內容生產中選擇一個 capstone。
- 必須包含 API、持久化、追蹤、20 例評測、一次故障注入和恢復。
- 為其餘項目各寫一頁 design review，說明不做什麼以及原因。

退出標準

- 展示端到端 demo、CI 報告、評測結果與成本表。
- 至少證明一個失敗路徑可恢復、一個安全邊界不可繞過。

校正提醒

語音項目應補 WER、端到端 P95、打斷時延、弱網恢復與人工轉接率；內容 / 影片項目應補採納率、平均修改次數、失敗恢復率與版權邊界。音色復刻必須獲得授權並提示 AI 身份。

R09

客服、銷售與內容智能體 / Customer Service, Sales and Content Agents

Cantonese recap, English explanation, example and key vocabulary

粵語速解 | Cantonese recap

客服、銷售同內容 Agent 唔係「駁個 LLM 就算」。每條流程都要有可靠資料來源、清楚狀態、逾時、重試、冚等
同人工接管。語音客服要表明係 AI，錄音同音色復刻要先獲授權；答唔到就拒答或轉真人。銷售考核可以輔助培
訓，但唔可以單靠模型分數決定獎懲。內容 Agent 只可用有授權來源，發佈前要做事實、版權同品牌審核。自進
化只代表提出新 Prompt 或策略候選，必須離線評測、人工批准、版本發布同可回滾，唔可以俾模型直接改線上規
則。成效要分場景量度，例如 P95 延遲、groundedness、採納率同成功任務成本。

English explanation

These agents need controlled workflows around the model. Reliability requires explicit states, timeouts, bounded retries, idempotency, and human escalation. Safety changes by use case: voice needs consent and handoff, sales scoring needs review and appeal, and content needs factual, copyright, and brand checks.

Example | 例子

A voice agent discloses that it is AI, retrieves the current approved price, and streams a grounded answer. Low retrieval confidence or user dispute triggers human transfer. Any proposed self-improvement runs in shadow mode against a frozen evaluation set before approval and versioned rollout with rollback.

Key vocabulary | 重點英文詞彙

English term	粵語意思 / Cantonese meaning
Human-in-the-loop	關鍵決定由人審核或批准
Groundedness	回答有冇根據指定資料來源
Barge-in	用戶講嘢時可即時打斷語音 AI
Circuit breaker	外部服務連續失敗時暫停呼叫
Escalation	將高風險或答唔到嘅個案轉真人
Policy gate	發佈前檢查安全、版權同業務規則

Revision tip | 溫習貼士

遮住右邊解釋，只睇 English term，用廣東話講一次意思，再用英文講一個例子。If you cannot explain it both ways, mark it for Day 1, Day 3 and Day 7 review.

Q09

Customer Service, Sales and Content Agents Revision Quiz

Do it closed-book first. Answers are collected near the end of the guide.

做法：先唔好揭答案。MCQ 每題 1 分，短答每題 2 分，實作題 3 分；滿分 10 分。Try to answer the short questions in both Cantonese and English.

1. What is the safest self-evolution workflow?
 - A. Rewrite production prompts immediately
 - B. Evaluate offline, obtain approval, and deploy with rollback
 - C. Turn every positive comment into a rule
 - D. Give the model administrator access
2. How should an LLM-generated sales assessment be used?
 - A. As the sole basis for dismissal
 - B. As a public ranking
 - C. As training feedback with review and appeal
 - D. As proof of misconduct
3. What should happen when a content provider repeatedly times out?
 - A. Retry forever
 - B. Publish incomplete output
 - C. Use bounded retries, a circuit breaker, and a safe fallback
 - D. Remove timeouts

Short answer | 短答

4. Why do voice agents need metrics beyond text-response accuracy?

5. What checks should run before AI-generated marketing content is published?

Mini practical challenge | 小實作

6. Design a five-node workflow turning an authorized customer question into reusable sales content, with retrieval, safety, approval, failure handling, and monitoring.

Your evidence / 你嘅證據：

Score guide

8-10: ready to move on. 6-7: review your weak items tomorrow. 0-5: re-run the smallest example before continuing.

10

簡歷與項目面試

把每句話變成可展示的證據

閱讀順序

- 項目4-面試題一問一答追問版 副本.pdf
- 項目部分知識點-面試題 副本.pdf
- 項目面試輔導（視頻版） 副本.pdf
- 5.0 8個項目面試題.pdf
- 6.0新增 面試力爆炸-將你的項目放大十倍.pdf
- 如何用Ai寫一份好的簡歷 && 優秀學生簡歷參考.pdf
- 懂王-Ai應用開發-獨家絕密押題200道 副本.pdf（診斷用，不線性通讀）

本週要掌握

- 用 STAR 講真實問題、行動、結果與限制。
- 區分個人項目、課程延伸、開源貢獻、客戶試點與生產經驗。
- 只使用有腳本、環境、樣本量和原始結果支持的指標。

動手交付

- 完成一頁架構圖、3 個 90 秒項目陳述、8 個證據化 STAR 故事。
- 做一次計時 30 題模擬面試，並把錯誤變成溫習卡。
- 寫一頁真實簡歷；每條項目 bullet 建立 evidence card。

退出標準

- 能現場打開代碼、運行測試或展示報告證明每項核心聲明。
- 對未上線內容能清楚說“已實現、已驗證、設計方案或後續計劃”。

校正提醒

不得虛構公司、年限、團隊、用戶量、營收或事故。課程源碼必須注明起始模板，並列出自己的新增模組、測試與提交記錄。沒有真實業務指標時，寫“本地基準”“模擬負載”或“目標 SLO”。

R10

履歷、面試證據與誠信 / CV, Interview Evidence and Integrity

Cantonese recap, English explanation, example and key vocabulary

粵語速解 | Cantonese recap

履歷最重要唔係寫到似大廠，而係每一句都證明到。公司、年資、團隊、人數、日活、收入同效能數字都唔可以作；課程起始碼亦要講明來源。最穩陣做法係把內容分成「已實作、已驗證、設計方案、下一步」，再為每個成果整 evidence card：程式位置、測試方法、環境、樣本量、原始結果同個人貢獻。冇正式上線，就寫「本地基準」或「雲端沙盒」；未做，只可以話「我會點設計」。面試用 STAR 結構都要講真，遇到唔識可以承認，再展示推理。Demo 仲要移除客戶資料、API key 同公司機密。誠信唔會削弱作品，反而令追問時更有底氣。

English explanation

A credible CV separates implementation, verification, design, and future work. Every numerical claim should connect to reproducible evidence, while course or open-source starting points should be acknowledged. A truthful statement with scope and limitations is stronger under follow-up than an impressive claim that cannot be demonstrated.

Example | 例子

Avoid 'Reduced production TTFT from 15 seconds to 2 seconds for 10,000 users' without records. Prefer: 'In a local 50-query benchmark, reduced median TTFT from 3.4 seconds to 2.1 seconds by shortening the fixed prompt; script and environment are documented.' If Redis failover was only designed, say proposed rather than implemented.

Key vocabulary | 重點英文詞彙

English term	粵語意思 / Cantonese meaning
Claim-evidence matrix	每項履歷主張都有對應證據
Reproducible benchmark	別人可以按相同條件重新測試
Scope of contribution	清楚交代自己實際負責邊部分
Design target	尚未實測或達成嘅預定目標
Provenance	程式碼、資料同素材嘅來源紀錄
Confidentiality	保護客戶、僱主同項目機密嘅責任

Revision tip | 溫習貼士

遮住右邊解釋，只睇 English term，用廣東話講一次意思，再用英文講一個例子。If you cannot explain it both ways, mark it for Day 1, Day 3 and Day 7 review.

Q10

CV, Interview Evidence and Integrity Revision Quiz

Do it closed-book first. Answers are collected near the end of the guide.

做法：先唔好揭答案。MCQ 每題 1 分，短答每題 2 分，實作題 3 分；滿分 10 分。Try to answer the short questions in both Cantonese and English.

- Which CV statement is most credible?
 - Built an enterprise platform for millions
 - Mastered every AI framework
 - Extended a course starter with tenant isolation and 18 tests
 - Led a five-person team when working alone
- You designed but did not implement a distributed queue. How should you describe it?
 - Deployed a production queue
 - Proposed and documented a Redis-based queue design
 - Owed the company platform
 - Omit the distinction
- Which item is safe to include in a public demo?
 - A live API key
 - Raw customer conversations
 - Synthetic data and redacted placeholders
 - An employer's private repository

Short answer | 短答

- What information should an evidence card contain?

- How should you answer about production users if the project was never deployed?

Mini practical challenge | 小實作

- Rewrite this ethically: 'Built a self-evolving sales Agent used by 20,000 daily users, improving conversion 35% and supporting 10,000 QPS' when only a local prototype and 100-case evaluation exist.

Your evidence / 你嘅證據：

Score guide

8-10: ready to move on. 6-7: review your weak items tomorrow. 0-5: re-run the smallest example before continuing.

新增內容 1：技術決策門

在引入 RAG、微調或 Agent 前先回答：問題到底是知識、行為，還是行動？

需求	優先方案	何時升級	最低評測
規則明確、必須確定	普通代碼/規則引擎	規則無法覆蓋語言變體時加 LLM	單元測試、邊界測試
知識外部且經常變化	RAG	需要跨源、多跳或動態工具時加 workflow/agent	Recall@k、MRR、忠實度、引用準確率
輸出風格、格式或工具行為反復失敗	Prompt + structured output	凍結測試集證明仍不足時再微調	任務成功率、schema 有效率、盲測
需要多步決策與外部動作	Workflow 優先，必要時 Agent	路徑不可預先枚舉時增加自主性	工具選擇、參數、軌跡、成本、越權率
穩定行為差距，且有授權高質數據	微調	收益超過訓練、部署與治理成本時採用	同集基線、安全、記憶洩漏、成本

簡單基線優先

每次升級複雜度都必須在同一凍結數據集上戰勝更簡單的基線。否則複雜架構只是在增加維護、成本與攻擊面。

新增內容 2：統一評測框架

層級	核心問題	推薦指標	證據
Retrieval	找對了嗎？	Recall@k、MRR/nDCG、ACL 違規率	凍結問答集與相關文檔標注
Generation	答對且有根據嗎？	正確性、忠實度、拒答準確率、引用準確率	參考答案、人工抽檢、bad cases
Agent	走對步驟了嗎？	任務成功率、工具選擇、參數有效率、軌跡長度、越權率	期望工具軌跡與審批日誌
System	穩定且付得起嗎？	P50/P95/P99、錯誤率、重試、隊列等待、Token、成功任務成本	trace、metric、log、成本報表
Safety	會洩漏或被操縱嗎？	注入成功率、敏感資料洩漏、跨租戶洩漏、毒化檢測率	紅隊集、權限測試、審計記錄
Human	用戶真的接受嗎？	採納率、修改次數、轉人工率、申訴一致性	匿名反饋與人工復核

評測閉環

- 用 10-20 個高質素例子起步，覆蓋常見路徑與邊界條件。
- 在部署前做 offline experiment；版本、Prompt、模型或索引改變都跑回歸。
- 在生產中做抽樣 online evaluation，並記錄延遲、成本、用戶反饋與安全事件。
- 把失敗 trace 轉成新的離線測試；LLM-as-judge 必須有人類抽檢和校準。
- 報告樣本量、數據日期、模型版本、置信區間或至少分布，不只展示一個平均值。

新增內容 3：生產級 Agent 發佈門

門	必須滿足	不滿足時
Scope	目標、非目標、工具白名單、讀寫等級明確	退回 workflow 或只讀模式
Identity	用戶、租戶、工具和數據權限均由確定性系統驗證	禁止依靠 system prompt 充當鑒權
Approval	寫入、刪除、支付、外發、權限變更有明確預覽與批准	阻斷調用
Reliability	超時、重試、冚等鍵、停止條件、預算、降級與回滾存在	僅限沙箱/測試
Security	提示注入、惡意檢索內容、RAG 投毒、工具返回與上傳文件均按不可信處理	修復後重跑紅隊集
Observability	trace/metric/log 關聯，敏感欄位脫敏，有告警與 runbook	不可生產發佈
Evaluation	凍結集通過，軌跡與工具參數達標，人工抽檢通過	保持 canary

MCP 重點

MCP 提供連接標準，不自動提供安全。遠程 server 需要最小權限、明確同意、審計、令牌受眾驗證和下游令牌隔離；工具結果、resources 與 prompts 都可能攜帶不可信內容。

新增內容 4：項目工程品質清單

維度	應補內容	可驗收證據
架構	MVP/生產邊界、ADR、狀態機、冚等、租約、失敗恢復	架構圖、ADR、failure matrix、恢復 demo
測試	unit、provider contract、DB integration、E2E、load、fault injection	CI 報告、測試目錄、壓測腳本
安全	密鑰、RBAC/ACL、webhook 驗簽、上傳校驗、注入/毒化、日誌脫敏	威脅模型、secret scan、安全報告
可觀測性	API、隊列、MCP、模型和 DB 的 trace/metric/log	dashboard、告警、runbook
成本	預算、併發、超時、分級路由、緩存命中率、成功任務成本	成本報表、模型路由實驗
評測	凍結集、簡單基線、回歸、人工校準、bad cases	版本化數據集、對照報告
治理	授權、用途、保留、刪除、來源、版權、模型/數據卡	審批記錄、刪除演練、license 清單

推薦項目指標

- 語音客服：WER、端到端 P95、打斷時延、弱網恢復率、人工轉接成功率。
- RAG/數據中台：Recall@k、MRR/nDCG、忠實度、拒答準確率、跨租戶洩漏率。
- 內容 / 影片：任務完成率、片段採納率、平均修改次數、失敗恢復率。
- 微調：與 base+prompt+RAG 同集對比，並測記憶洩漏、安全退化、延遲與成本。
- 銷售考核：AI 與多名人工評審的一致性；保留復核和申訴。
- 自進化 Agent：只產出候選，經離線評測、人工批准、版本發佈和回滾後才生效。

新增內容 5：求職誠信與證據卡

風險做法	更有說服力的替代
虛構公司、年限或團隊	寫明個人項目、課程延伸或開源貢獻；如實標注獨立完成與協作範圍
把規劃功能說成已上線	分成已實現、已驗證、設計方案、後續計劃四欄
使用預寫性能數字	跑可復現基準，寫清設備、數據集、流量模型、樣本量和日期
把課程源碼當原創	注明模板來源，列出自己的模組、測試、性能改進與 commit
虛構生產事故或客戶	實現獲授權的小樣例，或坦誠說明只完成方案與驗證
編造日活、營收、轉化率	改寫為本地基準、模擬負載或目標 SLO

Evidence card 模板

欄位	填寫內容
Claim	簡歷或面試中要說的結論
My contribution	本人具體負責的代碼、設計與決策
Artifact	代碼位置、commit、架構圖、報告或 demo
Reproduce	一條可運行的命令或明確步驟
Measurement	環境、數據、樣本量、日期、before/after
Limitations	適用邊界、未驗證項和下一步

38 份 PDF 的建議閱讀模式

“深讀”要求完成實驗；“配套”用於補充；“查閱”按問題定位；“診斷”在完成對應項目後用於閉卷測試。

#	文件	模式	建議
1	0基礎到Ai銜接部分.pdf	深讀	建立應用數據流
2	新版-懂王Ai-Python從入小白到精通.pdf	深讀	結合糾錯實驗
3	LLM大模型原理 (claude4.6 精簡版) .pdf	深讀	先建概念骨架
4	LLM模型原理.pdf	查閱	按主題深入
5	推理模型LLM Api (物料) .pdf	深讀	改為能力矩陣
6	Rag架構、向量原理.pdf	深讀	先做檢索評測
7	LangChain案例.pdf	實驗	重點是向量/RAG
8	你非懂不可GraphRag.pdf	配套	補 local/global/DRIFT
9	Ai Agent全貌概覽.pdf	深讀	區分 workflow/agent
10	Agent集群架構 副本.pdf	配套	補權限與失敗恢復
11	深入淺出DeepAgent.pdf	配套	理解 harness 邊界
12	(新版2.0) 深入淺出LangChain.pdf	深讀	作為框架主教材
13	LangChain全解析.pdf	配套	補 Agent API
14	深入淺出LangChain.pdf	對照	識別版本漂移
15	(2.0新版) 深入淺出LangGraph.pdf	深讀	作為工作流主教材
16	LangGraph全解析.pdf	配套	補狀態與持久化
17	LangSmith鏈路追蹤監控.pdf	深讀	擴展到 offline/online eval
18	數據標注平台搭建與實戰 副本.pdf	深讀	補授權和 QA
19	主流模型微調方案解讀 副本.pdf	深讀	把經驗法則改成實驗

38 份 PDF 的建議閱讀模式（續）

#	文件	模式	建議
20	LLaMA-Factory+Lora模型微調實戰 副本.pdf	實驗	跑小型 LoRA
21	項目3：聊天數據微調-完整流程文檔版本 副本.pdf	案例	重點看治理與洩漏
22	包裝電商項目需要下去瞭解的內容.pdf	掃讀	補真實業務流程
23	項目6：Ai數據中台.pdf	案例	數據血緣與品質
24	項目8：Ai素材管理中心.pdf	案例	多模態 ACL 與版權
25	項目9 商品交易自進化Agent -- Harness與Loop Engineer 架構設計.pdf	案例	候選-審批-發佈-回滾
26	項目一：Ai語音對話-智能客服項目 副本.pdf	案例	語音 SLO 與授權
27	項目5：直播切片Agent 項目.pdf	案例	任務恢復與採納率
28	項目二：自媒體運營Ai智能體 副本.pdf	案例	HITL 與版權
29	項目7：銷售考核Agent.pdf	案例	評審一致性與申訴
30	項目1：一問一答無限追問版 副本.pdf	診斷	構建語音項目後再讀
31	項目2：一問一答無限追問-面試題 副本.pdf	診斷	檢查 LangGraph 細節
32	項目4-面試題一問一答追問版 副本.pdf	診斷	MCP 架構閉卷測試
33	項目部分知識點-面試題 副本.pdf	診斷	跨項目查漏
34	項目面試輔導（視頻版） 副本.pdf	批判閱讀	保留技術，刪除造假建議
35	5.0 8個項目面試題.pdf	診斷	按項目抽題
36	6.0新增 面試力爆炸-將你的項目放大十倍.pdf	批判閱讀	只保留有證據的強化
37	如何用Ai寫一份好的簡歷 && 優秀學生簡歷參考.pdf	選讀	只學結構和表達
38	懂王-Ai應用開發-獨家絕密押題200道 副本.pdf	診斷	隨機抽題，不線性背誦

順序原則

先讀概念，再做對應功能，最後才使用面試問答檢查理解。文件名中的“新版”不代表永遠有效；涉及代碼的章節都要以鎖定版本的實驗為準。

Capstone 完成定義

第 9 週的項目只有同時滿足以下條件，才適合放入作品集。

- 可運行倉庫：README、版本鎖定、配置模板與一條啟動命令。
- 一頁架構圖：服務、數據、狀態、信任邊界和失敗恢復。
- 合法數據說明：來源、授權、去標識、保留與刪除。
- 自動化測試：unit、integration、E2E 與至少一次故障注入。
- 凍結評測集：基線、指標、bad cases、模型與數據版本。
- 安全說明：威脅模型、tool allowlist、審批點、secret scan、日誌脫敏。
- 可觀測性：端到端 trace、P95、錯誤率、重試、Token 與成功任務成本。
- 展示材料：3 分鐘 demo、架構說明、評測報告、evidence cards。
- 誠實邊界：明確個人貢獻、未上線項和限制。

最強的面試故事

不是“系統有很多高級名詞”，而是：我發現了一個可復現失敗，建立了基線，做了一個有控制變量的改動，用評測證明收益，同時解釋成本與邊界。

Revision Quiz Answer Key | 溫習測驗答案

點樣用答案

唔好只係對 A/B/C/D。錯咗要寫低點解其他選項唔啱；短答要對照 key points；實作題一定要有 code、test、diagram 或 report 做 evidence。

Topic 1: AI 與 Python 基礎 / AI and Python Foundations

- 1. B - Secrets belong on the server and should be injected through managed configuration, not exposed to the browser.
- 2. C - It keeps one project's packages and versions separate from other projects.
- 3. A - Validation prevents malformed or unsafe data from flowing into business logic and external services.

Short-answer key points

4. Browser sends HTTP/JSON; backend validates and applies rules; backend calls the LLM; backend returns safe JSON; frontend renders it.

5. Any two of: environment-based secrets, authentication, input validation, timeouts, safe logging, retries with limits, and explicit error handling.

Practical challenge checklist

- Valid input returns a successful JSON response.
- Invalid or blank input produces a validation error.
- No secret appears in source code, logs, tests, or responses.

Topic 2: LLM 原理與 API / LLM Principles and APIs

- 1. B - Inference uses learned parameters; it does not normally update them.
- 2. A - Lower temperature generally reduces randomness but does not guarantee correctness.
- 3. D - Schema enforcement plus validation is safer than trusting prose instructions alone.

Short-answer key points

4. Pre-training learns general next-token patterns; SFT teaches desired instruction behavior; inference generates output using fixed weights.

5. TTFT is time to the first token; total latency is time until the full response completes. Streaming mainly improves perceived responsiveness and TTFT.

Practical challenge checklist

- Both normal and streaming modes work for identical prompts.
- Every structured result is schema-validated.
- The report records model, temperature, TTFT, total latency, token use, and failures.

Topic 3: RAG 與 GraphRAG / RAG and GraphRAG

- 1. C - A large chunk can contain the answer plus substantial unrelated text, weakening the relevance signal.
- 2. B - A reranker applies a more precise relevance model to the initial candidate set.
- 3. D - Graph traversal is most useful when several entities and relationships must be connected.

Short-answer key points

4. Check ingestion, inspect chunks, confirm embedding compatibility, review metadata filters, test query wording and top-k, inspect candidates, then add reranking if evidence supports it.

5. Local search begins from relevant entities and nearby relationships; global search uses community-level summaries to answer broad whole-graph questions.

Practical challenge checklist

- The report includes hit@3 or Recall@k plus grounded-answer checks.
- Retrieved evidence is visible for every test question.
- The GraphRAG sketch names entities, relationships, and local/global routing.

Topic 4: Agent 與 MCP / Agents and MCP

- 1. B - The 2026-07-28 core is stateless and removed the old initialization and session requirements.
- 2. C - Security boundaries must be deterministic and enforced by the host or server.
- 3. D - Known sequences are easier to test and govern as workflows.

Short-answer key points

4. Tool output can be stale, wrong, malicious, or contain prompt injection. Validate schema and source, and never delegate authorization to text.

5. The server returns input_required; the client gathers the answer and retries the original request with inputResponses.

Practical challenge checklist

- Search and drafting may be automatic; submission requires authenticated approval.
- Writes use idempotency, bounded retries, call and cost limits, and audit logs.
- Supplier text is schema-validated and isolated from trusted instructions.

Topic 5: LangChain 應用 / LangChain Applications

- 1. A - create_agent is the primary high-level v1 agent interface.
- 2. B - Validated structured output is exposed through structured_response.
- 3. B - Middleware centralizes cross-cutting policy and lifecycle behavior.

Short-answer key points

4. ProviderStrategy uses native provider schema enforcement; ToolStrategy obtains structured data through tool calling when that path is appropriate.

5. Examples include authenticated user or tenant ID, permissions, database handles, feature flags, and request-scoped configuration.

Practical challenge checklist

- Use create_agent with read-only lookup_order and approval-gated refund_order.
- Put authenticated identity and permissions in runtime context.
- Use a typed response schema plus PII and human-approval middleware.

Topic 6: LangGraph 與 LangSmith / LangGraph and LangSmith

- 1. C - The node is replayed from its start, which is why pre-interrupt side effects need special care.
- 2. B - Persistence stores snapshots; the thread identifier selects the logical run or conversation.
- 3. A - Offline evaluation uses curated datasets before release; online evaluation samples deployed behavior.

Short-answer key points

4. Put the write in a separate node after approval, or make it idempotent with a stable operation key. Never assume pre-interrupt code runs only once.

5. Capture the trace, remove sensitive data, diagnose and label it, add it to a curated dataset, test the fix offline, then monitor the new version online.

Practical challenge checklist

- The write occurs after approval and uses a stable idempotency key.
 - The interrupt payload is JSON-serializable and the same thread_id resumes the run.
 - Offline tests cover routing and arguments; online checks cover safety, latency, or failures.
-

Topic 7: 微調與數據工程 / Fine-tuning and Data Engineering

- 1. B - Frequently changing external knowledge is better retrieved at request time.
- 2. C - Conversation-level splitting prevents near-duplicate turns from appearing on both sides of evaluation.
- 3. D - Training memory includes much more than the quantized base weights.

Short-answer key points

4. Dataset hash and split IDs, model and tokenizer revision, chat template, seed, hyperparameters, code and package versions, hardware, and evaluation definitions and results.

5. When reliable preference pairs exist and controlled evaluation shows that SFT still has a measurable preference, tone, or behavior gap.

Practical challenge checklist

- Authorization and PII handling happen before training preparation.
- Splitting is by conversation, customer, or time before sample expansion.
- Prompt, RAG, and SFT baselines are compared on quality, privacy, latency, and cost.

Topic 8: 數據、素材庫與電商專案 / Data, Assets and E-commerce Projects

- 1. B - Production data must pass authorization, policy, provenance, and quality controls.
- 2. C - Hard access and identity filters should narrow the candidate set before ranking.
- 3. C - Once exposed, a credential must be considered compromised even if the visible copy is removed.

Short-answer key points

4. Traceability supports replay and audit, but it never overrides consent, purpose limitation, retention periods, or deletion rights.

5. Examples: tenant-A queries against tenant-B documents, ACL-filter tests, deleted-document retrieval tests, and adversarial identifier tests.

Practical challenge checklist

- Use raw -> staging -> approved with consent and provenance.
- Apply tenant, permission, and SKU filtering before vector search; use retrieval text for media.
- Include safe fallback plus Recall@K, leakage rate, and cost per successful query.

Topic 9: 客服、銷售與內容智能體 / Customer Service, Sales and Content Agents

- 1. B - Changes need controlled evaluation, approval, versioning, and rollback.
- 2. C - High-impact people decisions require human review and a contestable process.
- 3. C - Bounded recovery prevents runaway latency and cost while preserving a controlled fallback.

Short-answer key points

4. Voice quality also depends on ASR accuracy, end-to-end latency, interruption response, packet loss, TTS quality, and successful human handoff.

5. Check factual grounding, consent, copyright and source rights, personal-data removal, brand policy, and platform compliance.

Practical challenge checklist

- A sound path is receive/consent -> authorized retrieval -> grounded draft -> policy gate -> human approval/publish.
- Add idempotency, timeout, bounded retry, escalation, and rollback.
- Track P95 latency, groundedness, task success, and human acceptance rate.

Topic 10: 履歷、面試證據與誠信 / CV, Interview Evidence and Integrity

- 1. C - It states the starting point and a specific, verifiable contribution.
- 2. B - The wording accurately separates design work from implementation and deployment.
- 3. C - Synthetic and redacted materials protect users, employers, and credentials.

Short-answer key points

4. Claim, personal contribution, code or commit, reproduction method, environment, dataset and sample size, raw result, limitations, and date.

5. Say clearly that it was not production-deployed, describe the verified local workload, and separate measured results from capacity targets.

Practical challenge checklist

- State that it is a local prototype, not a production deployment.
 - Mention the 100-case evaluation and the actually implemented approval/versioning features.
 - Describe scaling as a documented design target, not a measured achievement.
-

官方參考資料

用於校正版本漂移與補充生產標準。訪問日期：2026-09-03。

ID	來源	連結
R1	Python data model	https://docs.python.org/3/reference/datamodel.html
R2	Python free-threading HOWTO	https://docs.python.org/3/howto/free-threading-python.html
R3	uv project layout	https://docs.astral.sh/uv/concepts/projects/layout/
R4	LangChain: frameworks, runtimes, and harnesses	https://docs.langchain.com/oss/python/concepts/products
R5	LangGraph overview	https://docs.langchain.com/oss/python/langgraph/overview
R6	LangGraph interrupts	https://docs.langchain.com/oss/python/langgraph/interrupts
R7	LangSmith evaluation concepts	https://docs.langchain.com/langsmith/evaluation-concepts
R8	LangChain retrieval architectures	https://docs.langchain.com/oss/python/langchain/retrieval
R9	Microsoft GraphRAG query overview	https://microsoft.github.io/graphrag/query/overview/
R10	MCP 2026-07-28 specification release	https://blog.modelcontextprotocol.io/posts/2026-07-28/
R11	OWASP 2025 Top 10 for LLM and GenAI applications	https://genai.owasp.org/llm-top-10/
R12	NIST AI 600-1 Generative AI Profile	https://airc.nist.gov/technical-reports/
R13	OpenTelemetry observability primer	https://opentelemetry.io/docs/concepts/observability-primer/
R14	Kubernetes Secrets good practices	https://kubernetes.io/docs/concepts/security/secrets-good-practices/
R15	Hugging Face PEFT LoRA	https://huggingface.co/docs/peft/package_reference/lora
R16	Hugging Face TRL dataset formats	https://huggingface.co/docs/trl/en/dataset_formats

版本原則

框架、協議和模型介面變化很快。每章都應寫明測試日期、依賴版本、模型 revision 和供應商；更新前先跑契約測試與凍結評測集，不要把某一版文檔當成永久事實。

最後一頁：從今天開始

今天只做四件事，不要一次打開 38 份 PDF：

- 完成安全清零：憑據、數據授權、日誌與倉庫。
- 建立 study-lab 倉庫，寫下 Python/模型/框架版本並提交 lockfile。
- 讀 Week 1 的第一份 PDF，完成一頁學習卡。
- 在日曆安排本週的 2 次動手時段和第 1、3、7 天溫習。

本週承諾

我會完成：_____

完成證據：_____

截止日期：_____

閱讀資料讓你知道有哪些概念；可復現證據讓你證明自己真正掌握了它們。